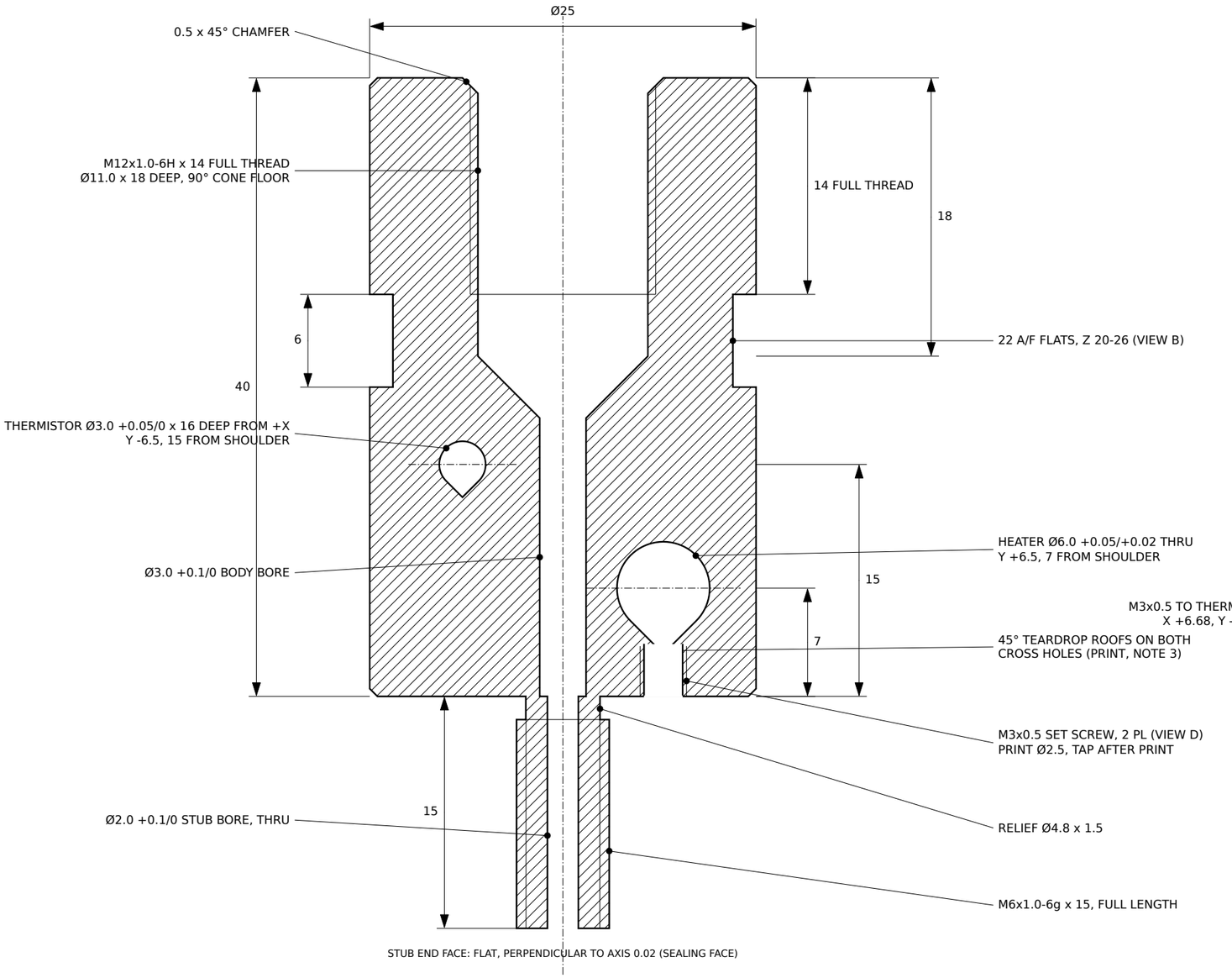
NOTES (UNITS: mm)

1. MATERIAL: 316L STAINLESS STEEL, METAL 3D PRINTED (SLM). NO ALUMINIUM (280 °C SERVICE).
2. GENERAL TOLERANCE ±0.1 UNLESS NOTED. BREAK SHARP EDGES 0.3-0.5. 0.3 x 45° CHAMFER ON HEATER AND THERMISTOR HOLE ENTRIES.
3. PRINT AXIS VERTICAL, TOP (M12) FACE ON THE PLATE, STUB UP. CROSS HOLES HAVE 45° TEARDROP ROOFS POINTING TO THE SHOULDER; M12 HOLE FLOOR IS A 90° CONE.
4. CAD: barrel_coupler.step HAS ALL THREADS MODELLED; barrel_coupler_nothreads.step HAS A PLAIN Ø6 STUB AND Ø11 / Ø2.5 HOLES.
5. AFTER STRESS RELIEF AND SUPPORT REMOVAL: FACE THE STUB END SQUARE, CHASE M6 (DIE) AND M12 (TAP), TAP BOTH M3x0.5 FROM Ø2.5, REAM HEATER HOLE IF UNDERSIZE.
6. CONCENTRICITY M12 BORE TO M6 STUB: 0.05 TIR.
7. BORE MUST NOT BREAK INTO EITHER CROSS HOLE (2.0 MIN WALL TO HEATER HOLE). STUB WALL 1.39 MIN AT THE M6 ROOT.
8. HEATER AND THERMISTOR HOLES RUN PARALLEL TO X, PERPENDICULAR TO THE AXIS, ON OPPOSITE SIDES OF IT.
9. DEBURR THE BORE BOTH ENDS. LEAVE Ø25 ROUND ABOVE THE FLATS FOR AN SK25 SHAFT CLAMP. CUSTOMER SUPPLIES M12x1.0 TAP IF NEEDED.
10. INSPECT: M6x1.0 NUT RUNS FULL STUB; M12x1.0 TAP OR GAUGE RUNS 14 DEEP; 6 mm CARTRIDGE SLIDES IN; CLEAR SIGHT LINE THROUGH THE BORE.
11. THREADS SHOWN BY CONVENTION. THIRD-ANGLE PROJECTION. WHERE THIS DRAWING AND shop_order.md DIFFER, THE SHOP ORDER GOVERNS.

SECTION A-A

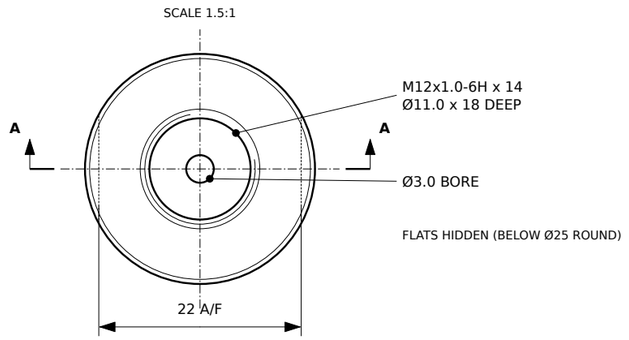
PLANE X = 0, VIEWED FROM +X SCALE 3:1



STUB END FACE: FLAT, PERPENDICULAR TO AXIS 0.02 (SEALING FACE)

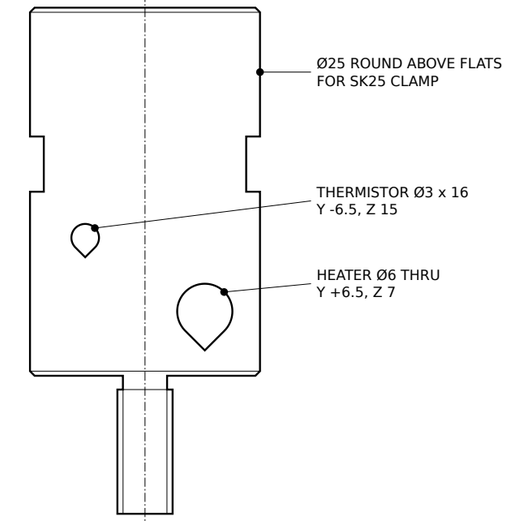
VIEW B (TOP, FROM +Z)

SCALE 1.5:1



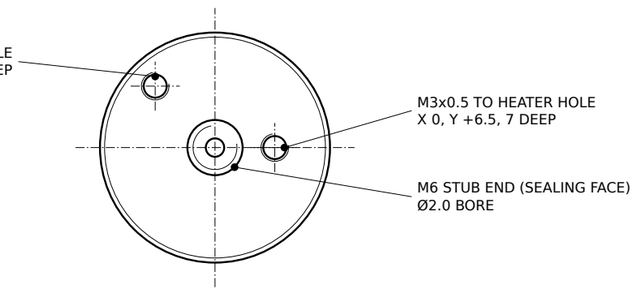
VIEW C (FROM +X)

SCALE 1.5:1



VIEW D (SHOULDER FACE, FROM -Z)

SCALE 1.5:1 +X UP



HEATED BARREL COUPLER	
316L STAINLESS (SLM) UNITS mm QTY 1	
CUSTOMER: ADAM HUMPHRIES REF: barrel_coupler.step	DWG: BC-001 REV B SCALE: AS NOTED SHEET 1/1